higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

QUANTITY SURVEYING N4

15 APRIL 2019

This marking guideline consists of 8 pages.

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SECTION A

QUESTION 1

1.1.1

2/2,000 7,000 × 7,000 / 4,000 × 4,000 / 11,000 × 11,000 m ✓

(1)

1.1.2

11,00 ✓
11,00 ✓

C.s. of all deb. and veg. mat. ✓

(3)

1.2.1

4 × 7,000 = 28,000
Less 4/0,330 = 1,320
26,680 m✓

(1)

1.2.2

0,850 + 0,230 = 1,080 m✓

(1)

1.2.3

26,68
0,65 ✓
1,08 ✓

Excav. in firm ground for s.t. n.e. 2 m dp. from g.l.✓

(3)

1.2.4

2 ✓

26,68
1,08 ✓

R.o.c. to sides of s.t. n.e. 1,5 m dp.✓

(2)

1.2.5

2 ✓

26,68
0,16 ✓
0,85

B/fll. to sides of 330 mm thk. bk. wl.✓

0,650 – 0,330 ÷ 2
= 0,160 m ✓

(3)

1.3.1

26,68
0,65 ✓
0,23 ✓

15 MPa mass conc. in fdn. ftg.✓

(3)

1.3.2

6,56
5,21
0,10 ✓
2,73
1,35
0,10 ✓

15 MPa mass conc. in s.b. incl. s.o. & c. hor. top surf.✓

7,000 – 2/0,220 = 6,560 m✓
5,650 – 2/0,220 = 5,210 m✓
3,170 – 2/0,220 = 2,730 m✓

(6)

1.4.1

26,68
0,93 ✓

330 mm Thk. fdn. wl. in 1:3 c.m. in stock bks. blt. in s.b.✓

0,850 + 0,075= 0,925 m ✓

(3)

1.4.2

28,00 ✓
0,23 ✓

E.o. Stock bks. for fcgs.✓

3/0,075 = 0,225 m✓

(4)

[30]

QUESTION 2

2.1

1.1.1

2/2,000 7,000 × 7,000 / 4,000 4,000 / 11,000 × 11,000 m

1.1.2

11,00		C.s. of all deb. & veg. mat.
11,00	121,00	

1.2.1

4 x 7,000 = 28,000
Less 4/0,330 = 1,320
26,680 m

1.2.2

0,850 + 0,230 = 1,080 m

1.2.3

26,68		Excav. in firm ground for
0,65		s.t. n.e. 2 m dp. from g.l.
1,08	18,73	

1.2.4

2

26,68		R.o.c. to sides of s.t. n.e.
1,08	57,63	1,5 m dp.

(1)

2

26,68		B/fll. to sides of 330 mm
0,16		thk. bk. wl.
0,85	7,26	

0,650 – 0,330 ÷ 2
= 0,160 m

1.3.1

26,68		15 MPa mass conc. in
0,65		fdn. ftg.
0,23		

1.3.2

6,56		15 MPa mass conc. in
5,21		s.b. incl. s.o. & c. hor.
0,10		top surf.
2,73		
1,35		7,000 – 2/0,220 = 6,560 m
0,10		5,650 – 2/0,220 = 5,210 m
		3,170 – 2/0,220 = 2,730 m

1.4.1

26,68		330 mm Thk. fdn. wl. In
0,93		1:3
		c.m. in stock bks. blt. in
		s.b.

0,850 + 0,075 = 0,925 m

1.4.2

28,00		E.o. Stock bks. for fcgs.
0,23		

3/0,075 = 0,225 m

(2)

(2½)

2.2

		SMALL DWELLING					
EARTHWORKS							
SITE CLEARANCE							
m²							
C.s. of all deb., veg. mat. &							
prep. for bldg. wrk.							
			m³				
121	1						
			B/fll. to sides of 330 mm fdn.				
121 m²			wl.				
EXCAVATIONS			7,26	2			
m³							
			7 m³				
Excav. in firm grnd. for s.t. n.e. 2 m dp. from g.l.							
18,73	1						
19 m³							
m²							
R.o.c. to sides of s.t. n.e. 1,5 m dp. from g.l.							
57,63	1						
58 m²							

(3½)

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2.3

	SMALL DWELLING				
	EARTHWORKS				
	<u>Site clearance</u>				
1.	Clear site of all debris, vegetation matter and rubbish and prepare site for building work.	m²			
	<u>Excavations</u>				
2.	Excavate in firm ground for surface trenches not more than 2 metres deep from ground level.	m³			
		m²			
3.	Risk of collapse to sides of surface trenches not more than 1,5 metres deep from ground level.	m³			
4.	Backfilling to sides of 330 mm foundation wall.				
	Carried to summary			R	

WORKING UP - MARK ALLOCATION

<u>Squaring</u>		
Check side casts. (ticks)	0,5	
Check dimensioning. (ticks)	0,5	
Square dimensions.	0,5	
Check squaring answers. (ticks)	0,5	
Transfers to abstract.	0,5	(2,5)
<u>Abstracting</u>		
Section of work	0,5	
Trades	0,5	
Units	0,5	
Page referencing	0,5	
Use of columns	0,5	
Reducing	0,5	
Transfers to bill	0,5	(3,5)
<u>Billing</u>		
Section of work	0,5	
Bill no.	0,5	
Trade	0,5	
Item no.	0,5	
Descriptions (No abbreviations)	0,5	
Units	0,5	
Quantities	0,5	
Summary	0,5	(4)

(4)

[10]

TOTAL SECTION A: 40

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SECTION B

QUESTION 3

MEASURING LIST

Door - Dr.
- Pt.
- Dr. lck.

Frame - Fr.
- Pt.

Adjustments

Deductions - Wl.
- pls.
- Pt.

Additions - S.b.
- Scd. (10 × 1) **[10]**

QUESTION 4

4.1 External 110 mm brick skin wall between and over legs of roof trusses

4.2 Items recorded for fixing only

4.3 Word used when reference is made to the item last measured

4.4 Work measured as fixed in position

4.5 An item manufactured outside the standard specifications

4.6 Word used to indicate faulty dimensions being crossed out

4.7 When ordinary brickwork is finished off as faced

4.8 Refers to levelling the top surface of a concrete surface and keeping it wet to allow for the hardening process

4.9 The adding of similar items during 'taking off'

4.10 The action during abstracting to arrive at a final quantity

(10 × 1) **[10]**

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QUESTION 5

5.1
5.1.1 Item
5.1.2 Cubic metre
5.1.3 Cubic metre
5.1.4 Square metre
5.1.5 Square metre

(5 × 1) (5)

5.2
- Work completed as measured
- Materials on site not built in
- Variations
- Additions
- Work done by nominated subcontractors and materials on site
- Day works to be priced (Any 5 × 1) (5)

[10]

QUESTION 6

6.1 Wall plate✓ laid in cement mortar✓ (2)
6.2 Wrought✓ South African pine✓ timber✓ (3)
6.3 Build in✓ in door frame✓ (2)
6.4 Clear site✓ of all debris✓ and vegetation matter✓ (3)

[10]

QUESTION 7

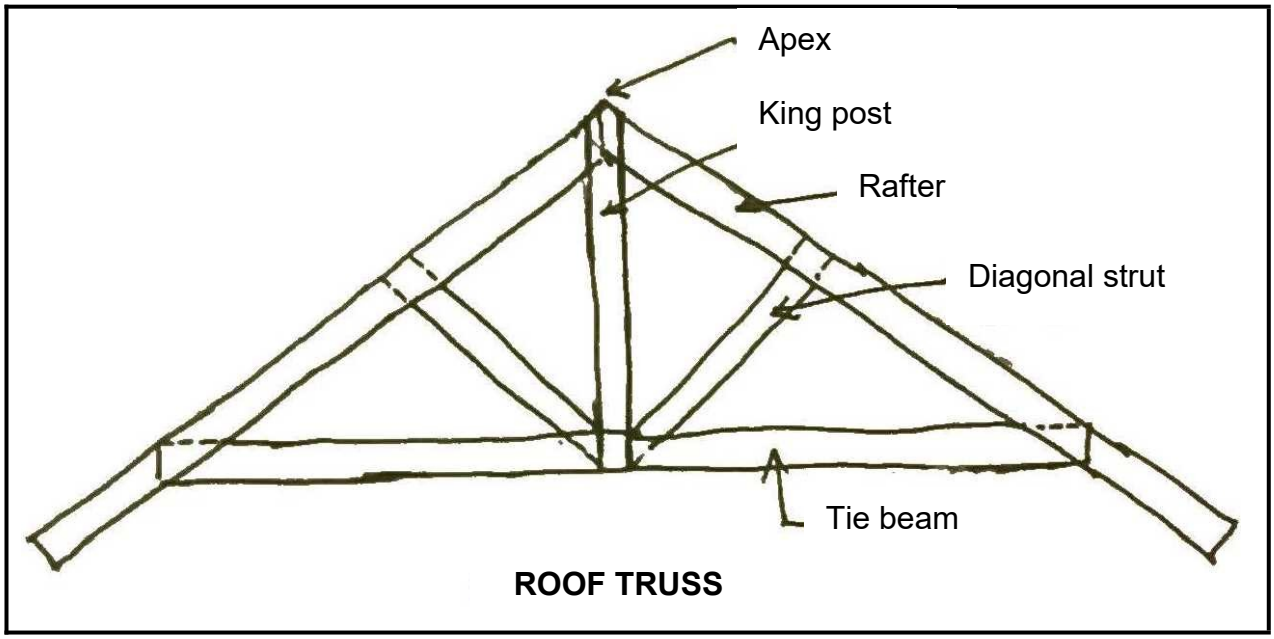
7.1
- Do cost estimates.
- Advise on selection of materials.
- Give advice on construction methods.
- Prepare contract documents.
- Write specifications.
- Call for tenders.
- Advise on type of tender. (Any 5 × 1) (5)

7.2
- Prepare sketched drawings.
- Check building site.
- Check boundary pegs.
- Check ground layers.
- Check site levels.
- Submit drawings for scrutiny.
- Advise clients on choice of quantity surveyor. (Any 5 × 1) (5)

[10]

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QUESTION 8



ROOF TRUSS

[10]

TOTAL SECTION B: 60
GRAND TOTAL: 100

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